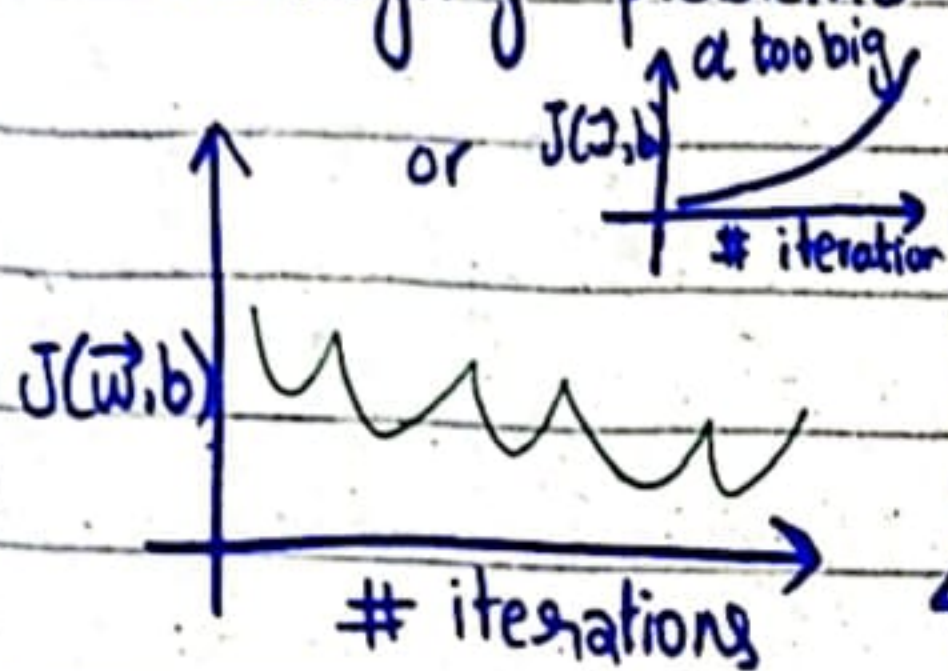


Choosing the learning Rate

if too small, run very slowly.

if too large, may never reach convergence.

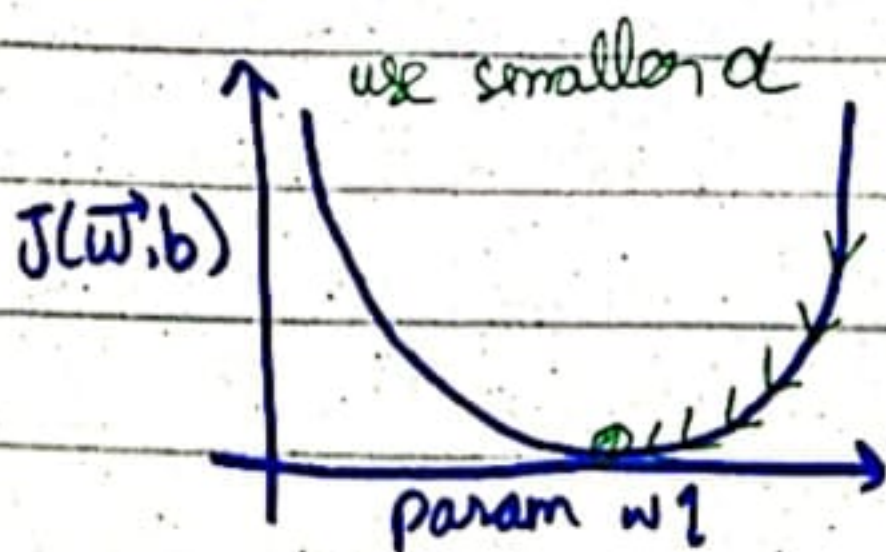
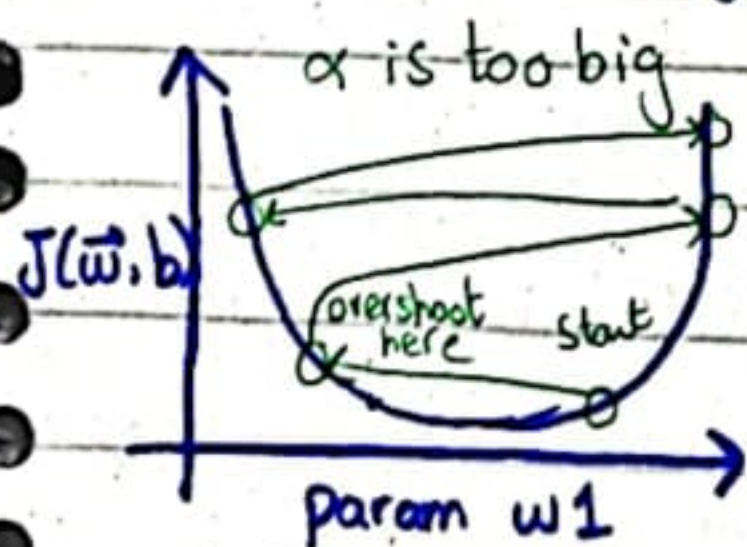
identifying problems with gradient descent



if you plot cost for no. of iterations, & notice that cost sometimes go up & sometimes go down.

→ Take that as sign gradient descent isn't working properly.

Adjust learning rate



Tip to see whether a bug or not:

choose ^{very} small α and see if value decreases on every iteration, if not there's a bug somewhere in the code.

→ only debug step not efficient for training model if alpha too too small.